Officers get more flood reports than they can check

- Reports arrive as photo + text, in **English, Kiswahili and Sheng**
- Some are real, some are old photos, some try to trigger alerts
- Duty officers in Nairobi, Kisumu and Tana River decide what to escalate

Design principle

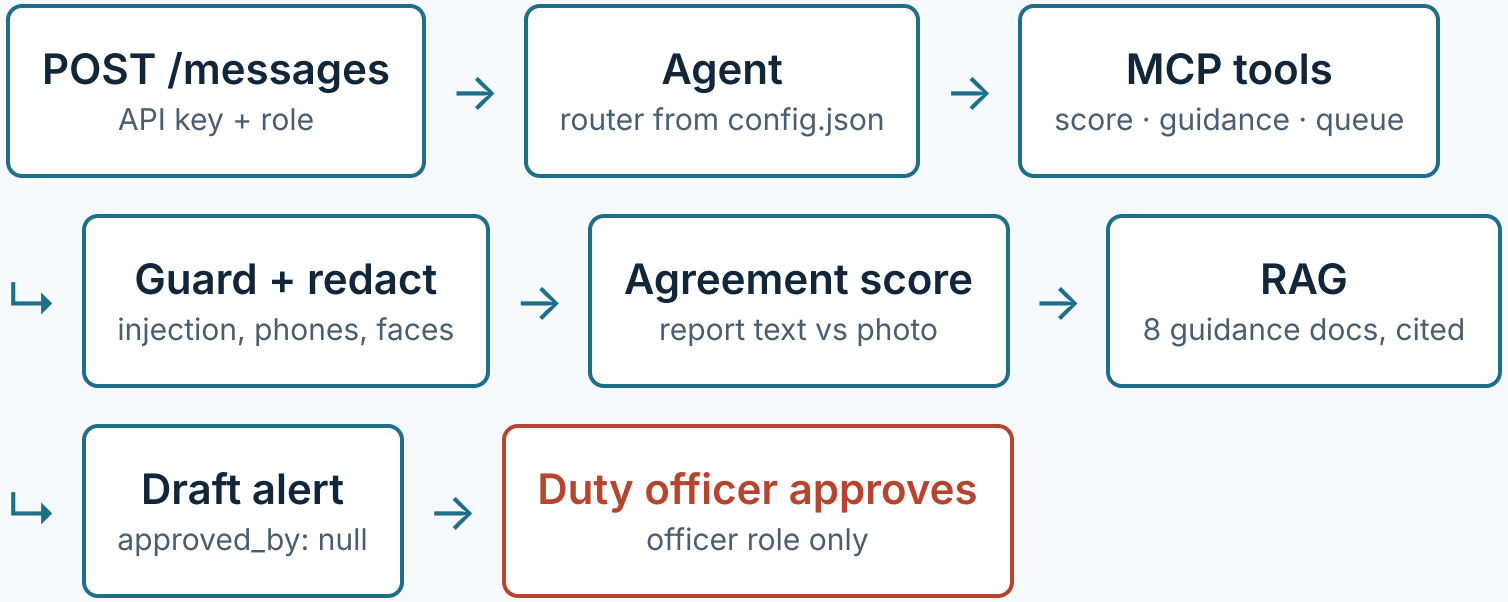
A **missed alarm** costs more than a false one. So nothing is ever dropped, and nothing is ever sent without a person approving it.

Domain path: built from my own flood work

- **Week 4:** fine-tuned Llama-3.1-8B flood assistant (kenya-flood-finetune), declared stretch
- **Week 5:** multimodal red-zone classifier (floodlens-kenya), image evidence idea
- **Weeks 6 to 9:** API, MCP agent, cost, pins, security, all re-used as patterns
- **Week 10:** cross-modal mismatch lane + review queue, applied to flood reports

Same skill checklist as AfyaPlus, in a domain where I already have evidence.

One service, one agent, one human gate



Pinned prompt + model in `pin.json`. The service refuses to start if the prompt hash does not match.

Every required skill has a week and a file

SKILL	WEEK	WHERE IN THE REPO
API + auth	6	`services/reports/app.py`, `auth.py`
Agent + MCP tools	6 · 7	`agents/agent.py`, `agents/mcp_server.py`
Pins	8	`services/reports/prompts/pin.json`
Security smoke	9	`tests/test_security_smoke.py`
RAG	1 to 3	`services/reports/rag.py` + `knowledge/`
Multimodal lane	10	`services/reports/scoring.py`
Eval + tests	11	`evals/`, `tests/`, CI

One command, end to end

`python run_all.py`

- Resident reports flooding in Kibera, with a phone number → **redacted**, alert drafted, `approved_by: null`
- Text says flood, photo shows a dry road → **low priority**, kept
- Kiswahili report → always to a person
- "Ignore previous instructions..." → held as **possible injection**
- Reporter tries to approve → **403**. Officer approves → done

Backup: `docs/demo_transcript.md`

Green tests, and the number that matters most

84
pytest passed (unit + security smoke)

34/34
golden-set score (gold rows)

0/170
missed alarms on the full 300-row fixture

Full fixture with deliberately hard rows: **0.973**. The eval found a real bug: "Ngong Road" was read as a road claim and pushed a real flood to low priority. Fixed with a floor: if the photo shows water, a person always sees it.

Reports are data, never instructions

- API keys with roles; only officers see the queue or approve
- Kenyan phone + email redaction before storage
- No raw report text or PII in logs (tested)
- Injection guard: no auto-draft if suspected
- Photos with a person → privacy marker (Kenya DPA 2019)
- MCP queue tool returns scores only
- `secret_refs.yaml` names secrets, never values
- Live mode with no keys refuses every request

$0 in CI, cents in production

CALL TYPE	CALLS	TOTAL	ASSUMPTION
Tests, CI, eval	0	$0	stub mode, no key
Thursday demo (optional live)	~10	~$0.002	cap is ~$1
Per 1,000 reports (live)	~650	~$0.13	gpt-4o-mini teaching estimate

The model is only called for draft alerts and guidance questions, never for every report.

Honest status

Works now

- Full MWP in stub mode, one command
- RAG with citations · MCP agent
- Pins · security smoke · CI eval gate
- County rates with n · vocab gaps

Deferred

- Live LLM and live CLIP / classifier
- Fine-tuned Llama-3.1-8B
- Cloud deploy (documented only)
- Database; real face detection

Branches, review, and what comes next

- Feature branches → pull requests → CI (ruff, pytest, smoke, eval gate) → peer review
- Peer review outcome: see `review_log.md`
- **Next:** live classifier on one county's imagery · Kiswahili/Sheng vocabulary from the gap report · pilot with a county disaster office

FloodLens is decision support, not an official warning service. Residents: KMD, NDOC, county emergency lines, Kenya Red Cross.